Xue Yufeng

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Education

Singapore University of Technology and Design Singapore, SG 487372

Bachelor of Computer Science First Class Honours *Sep. 2023 – Present*

4.52/5.00 GPA

Experience

Software Developer Intern May 2026 – Sep 2026

pQCee *Singapore, SG 079903*

- Made a generator that can create a quantum circuit to break ECC in cubic time whereas standard algorithms take significantly longer at exponential time. This was highlighted in a QCVE blog post and my former boss's Linkedin post
- Set up a pipeline to categorize 4000 requests per month to deliver to a client that had a net income of S$339 million in 2025.

FPGA mmWave Research Intern Sep 2025 – Dec 2025

Panasonic R&D Center *Singapore, SG 469332*

- Processed FMCW signals into a dataset for Neural Network prediction task, achieving MAE as low as 2.82 and 1.82, where the average error are as high as 9.9 and 6.9.
- Refactored 10+ fragmented entry points into a unified configuration file, reducing onboarding time for new developers and accelerating feature iteration cycles.
- Built a SQL data storage layer handling 10,000+ records and developed real-time and offline visualization pipelines to support model evaluation and experiment tracking.

Projects (Personal)

Microcontroller May 2026 - Sep 2026

- Implemented the whole OSI model on a microcontroller to be able to deliver 5 messages per second over UART.

Agendaler Website Feb 2026 – Mar 2026

- Protected client's S$50 of AI credits by restricting endpoints on Next.js server to authenticated users and implementing rate limiting and quota limiting, with reCaptcha considered.

EDA Banking System Nov 2025 – Present

- Integrated a microservice into a Kafka system tested with 100000 requests/second.

FPGA Game Jan 2025 – Apr 2025

- Optimized FPGA power consumption by 6x by implementing time-division multiplexing across 6 displays, replacing a naive always-on approach.

Ascenda Booking Platform May 2025 – Aug 2025

- Optimized autocomplete search latency by 85%+ (from 200ms to under 30ms) across a dataset of 70,000 records by implementing an efficient indexing strategy, significantly improving user experience.
- Set up an automatic CI/CD pipeline to test code, reducing 5 minutes of manual time per push down to 0 for a team of 8.
- Secured 10+ API endpoints using session authentication, preventing unauthorized access.
- Built a custom test reporter and mock dependency layer covering 50+ test cases, improving test visibility and reducing debugging time for the team.

SimpleDB May 2025 – Aug 2025

- Implemented a 2PL concurrency control mechanism, supporting up to 1,000 concurrent requests with deadlock detection and avoidance.

ParcelEye AI Tracker Jan 2025 – Apr 2025

- Deployed a real-time AI object tracking system capable of monitoring up to 100 simultaneous objects in a livestream, enabling scalable live surveillance use cases.
- Containerized the AI and reduced its image size from 32GB to 5GB by 84% using multi-stage builds.

Cat Dog Classifier Dec 2024 – Dec 2024

- Applied architectural features from proven CNN models, achieving 95% training and validation accuracy for binary classification.

Resume Website Oct 2024 – Present

- Developed an FFmpeg script to convert images to .webp, reducing total size of all images from 53500kB to 41500kB by 22%
- Migrated from CRACO to Vite for improved RAM usage by over 2x from a few GB to less than a GB during development.

Technical Skills

Languages: C, Python, Java, TypeScript, SQL
Frameworks & Libraries: Docker, FPGA (Lucid), Node.js, PyTorch, React, Spring Boot, SQL Databases, TensorFlow
DevOps & Tools: CI/CD, Docker, GitHub Actions
Cloud/Infrastructure: AWS, Firebase, GCP